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Startup founders and their LinkedIn connections: Are well-connected entrepreneurs more successful?

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ABSTRACT

For the past 40 years, entrepreneurs and researchers have assumed that entrepreneur networks are important for startup ventures. This study takes this notion further by testing whether these benefits translate into tangible financial outcomes for a startup. For this purpose, the study integrates two extensive databases that have not been studied together in previous research: Crunchbase.com, which provides information on the financial success of startup companies, and LinkedIn, which provides social network information of founders. The analysis revealed that several variables in LinkedIn profiles were positively correlated with the amount of funds raised by startup companies establishing a link between social networks and entrepreneurial success. The average number of followers that the founders of a company had according to their LinkedIn profile was the strongest predictor of the amount of funds raised by companies.

1. Introduction

Over the last 40 years, researchers in the social sciences have argued the importance of social networks in the entrepreneurial process (e.g., see Cooper & Bruno, 1977). In a theoretical essay, Dubini and Aldrich (1991) proposed the idea that entrepreneurs who are better connected through interpersonal networks have better access to valuable resources, which in turn, increases the likelihood of the success of a venture. While it is intuitive that the density of a founder's social network affects a startup's success, the relationship between social connections of a founder and tangible financial outcomes has not been systematically studied. We set out to fill this gap in the literature and explore the following questions: Are social networks of start-up founders related to their success? Specifically, is the quantity of connections of start-up founders associated with the amount of investment funds their companies have raised? Does it make sense for startup founders to invest time and effort in building networks? We aimed to investigate these questions building on the four-decade assertion that entrepreneurs are embedded in social networks and that these networks play a critical role in the entrepreneurial process (Aldrich & Zimmer, 1986).

There are at least two major reasons why such a relationship between founder networks and the amount of funds that are raised by companies is to be expected. First, well-connected founders and organizations have access to valuable resources such as knowledge,

expertise, human resource, and market information (see Davidsson & Honig, 2003; Hoang & Antoncic, 2003; Larson, 1991). Second, there is research suggesting that venture investors take founder connections into consideration while choosing to invest in one venture over another (Fried & Hisrich, 1994).

Despite reasons to expect a relationship between founder networks and the ability of startups to attract investment funds, this relationship has not been systematically studied. Network researchers have traditionally relied on self-reported ego-network data that limit the generalizability of the findings (see Hoang & Antoncic, 2003; Hoang & Yi, 2015). Recent databases, such as social networking site LinkedIn and crowdsourced startup database like Crunchbase, offer broader data sets that can be used to access founder connections and information on finances of startup companies, respectively. Beyond information on an aggregate level, both databases offer information on the level of individual founders and individual startup companies which is necessary to test whether there is an association between LinkedIn connections and entrepreneurial success.

We set out to achieve two major goals: First, we sought to empirically test the hypothesis that connections of startup founders are systematically related to tangible financial outcomes of companies. To this end, we sought to utilize data available in social media networking site LinkedIn and to explore if LinkedIn founder profiles are systematically associated with start-up success operationalized through the amount of funds raised and described in the database Crunchbase. For this study

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we looked at companies that were active in the information and technology industry and were located in the United States. The reason to restrict the industry and the geographic location of the companies and founders was to control for the effects that industry market size and location have on financial outcomes (see Zacharakis & Meyer, 2000, also Kakati, 2003).

Second, the study aimed to explore if potential connectedness of startup founders has positive impact on financial outcomes across three categories of startups ranging from those that are currently operating, those that have been acquired by other companies and those that have listed their companies on the stock exchange and have gone to initial public offering (IPO). Thus, we were also interested in finding out if a potential correlation between founder networks and investment funds acquired by a company can be observed across various stages of the entrepreneurial process.

In the following sections, research on entrepreneurial success and networking is described and the network hypothesis of entrepreneurial success is introduced. This is followed by a description of LinkedIn and Crunchbase and a demonstration of how these databases can be used to test the network hypothesis.

1.1. Entrepreneur networks and startup success

The idea that social networks are important for starting and sustaining new ventures is intuitive. Empirical research has shown that entrepreneurs do benefit from their social networks in multifarious ways. Moreover, founder networks are often used by funding agencies as predictors of startup success. Dubini and Aldrich (1991) theorized that better connected entrepreneurs have interpersonal connections which yield valuable resources and contribute to successful ventures. There is also empirical evidence that well-connected founders and organizations have access to valuable resources, albeit these resources are not direct financial outcomes but allied intangible outcomes. Greve and Salaff (2003, p.4) call such outcomes “complementary resources” and argue that networks of founders are crucial for companies to acquire and sustain these resources.

Accordingly, it has been shown that networks of founders can be helpful in seeking information and opportunities (Burt, 2000; Larson, 1991), acquiring competitive abilities (McEvily & Zaheer, 1999), influencing nascent entrepreneurship (Davidsson & Honig, 2003), and in establishing legitimacy by promoting reputation and credibility in the market or industry (Klyver, Hindle, & Meyer, 2008; also see; Hayter, 2016). For example, Larson (1991) conducted a case study of four entrepreneurial companies and looked at their inter-organizational connections. Based on these cases, Larson (1991) concluded that companies with more connections had greater levels of information exchange. This resulted in improvements in how the companies conducted their administrative work and helped make their customer response time more efficient. Some research suggests that connections to venture capitalists are particularly important as venture capitalists make the critical investment decisions—direct and indirect connections with venture capitalists increase the chance that a company is funded (Shane & Cable, 2002). Additional studies have shown that networks can provide startup ventures with reputational legitimacy (Klyver et al., 2008; Shane & Cable, 2002).

Taken together, this research suggests that startup companies with well-connected founders have access to more resources. A second reason why startups with well-connected founders may be more successful than startups founded by less-connected entrepreneurs is based on empirical research indicating that investors pay attention to the social networks of entrepreneurs. For example, Fried and Hisrich (1994) analyzed 18 successful start-up proposals which got funding from venture capitalists and concluded that if venture capitalists have had heard about the founder or the company through their networks, they valued the proposal higher than those of unknown companies and founders. Roure and Maidique (1986) did semi-structured interviews

with eight venture capitalists and concluded that prior experience of founders and their managerial capabilities were more reliable predictors of startup success than mere financial and technical criteria such as the viability of product and growth of target market. Moreover, founders who were able to recruit relevant people through their networks and establish more complete teams were likely to succeed more than startups with incomplete teams (also see Sandberg & Hofer, 1988, and Kakati, 2003).

Each of the described studies stresses the importance of network connections of founders for the success of startup companies. However, the studies did not provide any direct evidence for the claim that the number of connections of a founder is related to tangible financial outcomes. Some studies that stand out are studies conducted by Light (1984), Zimmer and Aldrich (1987), and Bates (1997). Light (1984) looked at ethnic minorities and how they acquire valuable resources through their networks to sustain their businesses. Light (1984) demonstrated that ethnic minorities mobilize financial and cultural resources through their interpersonal networks which is important for establishing and sustaining a small business venture. Zimmer and Aldrich (1987) further conducted 580 interviews of entrepreneurs and inquired how family and friends influence establishing a new venture, help sustain the business, and have a positive impact on turnover. They concluded that social ties have a positive and important influence on all three aspects of entrepreneurship. Bates (1997) surveyed 2,000 Chinese and Korean firms and explored how they raised money for their businesses and found that interpersonal ties were instrumental in providing access to financial resources for these firms. While these studies do provide evidence about the beneficial impact of social networks on financial resources, all the cited studies have been conducted with ethnic minorities. Also, as is evident from the cited literature, most of the research on entrepreneurial networks have adopted interviews, case studies or survey questionnaires as methods of data collection. While these methods provide insight and empirical evidence regarding the entrepreneurial process (see Elfring & Hulsink, 2003, also see Larson, 1991), the limited sample sizes make generalizability of the findings difficult. Moreover, research on entrepreneurial networks has traditionally relied on self-reported ego network data (Hoang & Antoncic, 2003) which is often limited in its scope and accuracy.

Part of the reason why the direct connection between founders' social networks and financial outcomes of a startup remain under-explored has been the absence of data sources that could provide information about founder networks. Such an absence of information sources had made empirical exploration of the network-outcome link a challenge. However, researchers now have access to social networking site LinkedIn and comprehensive start-up databases like Crunchbase.com, which can be used to access founder connections and information on finances of startup companies, respectively. This is in line with research arguing that cyber resources have enabled new processes of “interactions and interconnections among individuals, documents, data, analytic tools and concepts” (Huang, Contractor, & Yao, 2008, p. 28). Online communities like LinkedIn and Crunchbase.com are large-scale knowledge networks that contain multi-dimensional information which can be used to study networks and their role in different organizational contexts (Huang et al., 2008).

There is emerging evidence that online social networks are associated with funds raised in certain entrepreneurial contexts like crowdfunded enterprises. For example, Bellefamme, Lambert and Schwenbacher (2013) described various ways in which entrepreneurs seek funding through crowdfunding websites. Also, Mollick (2014) showed that online personal networks of owners (Facebook friends) of crowdfunded enterprises were positively associated with the amount of funds companies raised on the internet. However, crowdfunded startups and venture-capital backed startups differ in their organizational structure, funding needs, and network properties (Bellefamme et al., 2013; Mollick, 2014). Crowdfunded enterprises look for smaller amounts of investments from strangers on the internet. They do not

seek to influence sophisticated investors like venture capitalists. This study looks at founders of venture-capital backed startups to investigate whether founder social networks are associated with entrepreneurs' funds from sophisticated investors.

1.2. LinkedIn founder profiles and startup success

LinkedIn was launched in 2003 and currently has 467 million members across the world (LinkedIn, 2017, retrieved from <https://press.linkedin.com/about-linkedin>). Targeted at professionals to connect with other professionals and create a human resource pool, LinkedIn's profile, unlike that of other social networking sites, is like an online resume. LinkedIn provides information about a user's professional and educational background. Being a social networking site, LinkedIn also provides network cues that can be used to infer information about a founders' network.

For example, publicly accessible LinkedIn profiles provide information about the number of people that follow an individual. These ties can be used to make inferences about network size. LinkedIn data go beyond self-reported network data and provide a more objective resource of network information than self-reports. Within the entrepreneurial context, LinkedIn has only been used by very few studies.

There is limited understanding of how well LinkedIn represents the social networks of a person. However, researchers in organizational contexts like corporate recruitment have shown that LinkedIn is used by recruiters to get more information about potential applicants with 70% of the recruiters relying on LinkedIn to decide which applicants should be interviewed (Caers & Cateleys, 2011). Guillory and Hancock (2012) further showed that people are less likely to be deceptive in the information they provide over LinkedIn, particularly related to past work experience and skills compared to offline interactions and traditional resumes. In addition, Song and Vinig (2012) have done a pilot study of 100 entrepreneurs on LinkedIn and analyzed their network size and participation in diverse industries and showed that LinkedIn network size did have a positive relationship with entrepreneurial survival. The authors measured survival by looking if a company was able to survive over 3 years. However, they did not look at whether the connections were associated with financial outcomes of the companies. These studies indicate that there is merit in exploring if the network on LinkedIn can be used to test if founder connections are associated with start-up financial outcomes.

LinkedIn provides information about social networks but does not provide any information about the success of companies. Information on startup companies is provided by the comprehensive crowdsourced database Crunchbase.com. Crunchbase.com gathers information about all existing start-ups, their founders, current status, venture capital and other funding they have acquired through official statements of companies and investors. In doing so, they seek to ensure the credibility of the data that they collect. The crowdsourced database has been used in the fields of economics and management to study various financial aspects of the entrepreneurial process. For example, Crunchbase.com has been used to study venture capital funding in US internet startups during the financial crisis (Block & Sandner, 2009), and the database has also been used to study venture capital funding in the music industry (Waldner, Zsifkovits, & Heidenberger, 2012). Both studies have argued that [Crunchbase](http://Crunchbase.com) is a reliable and comprehensive source of startup ventures and investor funding. This study sought to combine the two information sources to explore if social connections on the social networking site LinkedIn are associated with tangible financial outcomes as reported on Crunchbase.com.

Specifically, we utilized the data available on LinkedIn and [Crunchbase](http://Crunchbase.com) to test if there is a positive relationship between start-up founder followers on LinkedIn and the funds raised by start-up ventures.

H1. Network Hypothesis. *The number of LinkedIn followers of founders is*

positively associated with the amount of money raised by companies per year.

The social network of founders is only one potential predictor that has been suggested in the literature. To control for individual founder characteristics such as professional experience that do not describe characteristics of social networks, additional variables provided on LinkedIn were included in the study. Professional experience, for the purposes of this study, was measured as the number of years an entrepreneur has been professionally active, the number of US states the entrepreneur has worked in, and the number of skills of an entrepreneur. This is in accordance with research by Sandberg and Hofer (1988) and Kakati (2003) who have shown that entrepreneurs' past experiences and track records positively impact startup performance. Using the information on LinkedIn that indicates entrepreneur experience, our second hypothesis proposed that founder experience is associated with funds raised by a startup.

H2. *The amount of experience of founders, as indicated on LinkedIn, is positively associated with the amount of money raised by companies per year.*

Furthermore, human capital (e.g., founder experience) and social capital indicators (e.g., founder network connections) have often been argued to be systematically related to each other (see Glaesar, Laibson & Sacerdote, 2002). On the other hand, other researchers have claimed that social networks and human capital indicators yield different kinds of benefits to an entrepreneurial venture where one cannot compensate for another (Davidsson & Honig, 2003). This informed our first two research questions.

RQ1. *Are the number of followers and the amount of experience of founders systematically correlated?*

And RQ2. *Does adding the number of founder followers as a predictor improve the prediction of the amount of funds raised by a company beyond founder expertise?*

Moreover, we were also interested in finding out if the proposed relationships hold across different phases of startups as well as within different phases by differentiating among three categories of startups. First are those that are currently in operation (*operating companies*). These are generally the youngest of the startups and those that are in their initial phases of operation and actively looking for funding to sustain themselves. Second are the ones that have been acquired by bigger companies (*acquired companies*). These were expected to be older than operating companies and to have more funding. The third category includes start-ups that have made it to the initial public offering (*IPO companies*). These companies were expected to be the oldest and the most successful startups. This informed our third research question.

RQ3. *Does the relationship between founder followers of companies and funds raised by companies differ across operating, acquired and IPO companies?*

In addition, we measured the companies' numbers of founders and the age of companies as both were expected to be directly related to the amount of funds received.

2. Method

To test the hypotheses, we analyzed data from Crunchbase.com and LinkedIn. There were two units of analysis in this study, the startup companies and their founders. First, we downloaded the data from startup database Crunchbase.com. Crunchbase.com is a crowdsourced online database of startups, founders, venture capitalists and other allied funding agencies. The data in Crunchbase.com is crowdsourced from startups, venture capital investors, funding agencies and media

reports. After randomly selecting companies, we systematically recorded the LinkedIn profiles of founders.

2.1. Sample

In the first step, the entire database of [Crunchbase.com](https://www.crunchbase.com) was downloaded after receiving permission from the website administrators. The initial data file had the name of all the companies in the database, the years they had been in operation, to which of the three categories they belonged (operating, acquired, or IPO), the number of rounds of funding they had received, and the amount of funds they had raised. In the second step, we selected the companies that were located in the United States and were operating in the information and technology industry. Market factors and industry characteristics have been shown to impact financial outcomes of startup companies (Zacharakis & Meyer, 2000, also Kakati, 2003). Our focus is to understand founder characteristics and by restricting the industry in which companies operate, we wanted to reach a relatively homogenous set of companies. We selected companies within the information and technology industry because it is one of the most actively invested industries in the United States (Huang, Wu, Lee, Bao, Hudson, & Boile, 2017). The selection procedure yielded 3,636 companies of which 3,060 were operating companies, 521 companies were acquired and 55 companies were IPOs. In the third step, we randomly selected 50 companies from each of the three category that were then used for further analyses.

For each of the 150 selected companies, we gathered information from [Crunchbase](https://www.crunchbase.com) about the founders of the respective companies. It turned out that [Crunchbase.com](https://www.crunchbase.com) did not have complete information about founders for all the randomly selected companies, particularly for the companies that have gone to IPO. Therefore, we randomly selected more companies from the original database until we had 50 companies in each of the categories that had complete information. We found 50 companies with complete information for operating and acquired companies, respectively. However, as there were only 55 companies with IPO status total, we could only collect complete founder information for 29 IPO companies.

This procedure yielded 129 companies that had a total of 227 founders. Some of the companies had single founders while others had multiple founders. After the names of the founders were collected, we accessed the public LinkedIn profile of each founder and recorded data about the number of connections of each founder, the year the founder started his/her professional career, the number of states the founder had been active in (worked in or received education in), and the number of skills of each founder.

2.2. Measures

Network Size. For the network size, we recorded the number of followers listed for each founder on the LinkedIn page. This is a number listed on the LinkedIn page. We took the number directly from the profile of the founder. Followers constitute connections of an individual and other people who might not be connections, but have been approved by the founder to have access to their posts and other information on their profiles.

Experience. LinkedIn provides several pieces of information about a founder on their public page. Three data points were recorded as indicators of the experience of a founder. First, the number of years a founder has had an active professional career. This is self-reported data that founders provide on LinkedIn. We recorded the year in which a founder entered workforce and calculated the number of years the founder had been professionally active until 2015.

As a second measure of experience, we recorded the number of US states a founder had either worked in or received a degree in. For example, if a founder was located in California and had worked in California throughout his/her professional life, then the founder had a “1” for this component of experience. However, if the founder had

institutional affiliation with more states than the one in which they were currently located, that number of states they had lived in was counted towards this component of experience. The geographic reach of the founder is considered as a measure of founder's work experience based on research that has shown that founders with wider geographic reach receive more informal training from communities of practice (Boud & Middleton, 2003) and tend to have higher managerial and social competence (Zahara, Ireland, Gutierrez & Hitt, 2000; Cummings, 2004; Lu & Beamish, 2004).

Third, we recorded data on the number of skills that were listed on a founder's LinkedIn profile. Skills on LinkedIn are self-reported. For the purpose of this study, we used the total number of skills that a founder had self-reported on their LinkedIn profiles, independent of whether the skill was endorsed by followers or not. Some examples of the skills listed were managerial skills, mergers and acquisition, strategic communication, and enterprise software.

Funds Raised and Age of Companies. We acquired the data on the funds raised by the company on [Crunchbase.com](https://www.crunchbase.com). This data was directly taken from the database. The funds constituted the money raised by the company through venture capitalist investors. The funds raised were divided by the age of the company to compute a measure of funds raised per year. This was done to control for the age of the company on funds as older companies are likely to have more funds. The age of companies was also provided on [Crunchbase](https://www.crunchbase.com).

3. Results

Table 1 provides descriptive statistics of the sampled companies that refer to [Crunchbase](https://www.crunchbase.com), and **Table 2** summarizes information on the founders of those companies that was taken from their LinkedIn profiles. As **Table 1** indicates, the companies ($N = 129$) raised an average of \$2.1 million each year ($SD = \$3,190,314.3$) and had on average 1.7 founders ($SD = 0.73$). The average number of rounds of investment per company was 2.42 ($SD = 1.71$), and the mean age of companies was 7.68 years ($SD = 5.97$).

The 129 companies were founded by 227 founders. As **Table 2** indicates, on average, the 227 founders had 1,146.68 followers ($SD = 1,125.13$), had worked in 2.89 US states ($SD = 2.08$), had an average of 21.69 skills ($SD = 14.76$), and had worked in the computer industry for an average of 16.99 years ($SD = 8.69$).

3.1. Founder LinkedIn connections and startup success

The network hypothesis (Hypothesis 1) predicted that the number of founder LinkedIn followers of a company is positively correlated with the amount of funds raised by the company per year.

In the dataset, funds were reported on the company level while the number of followers and founder experience is a characteristic of an individual founder. Most of the companies in the data set had more than one founder. As a measure of the social connectedness of a company, we averaged the number of followers of the founders of a company. For example, if a company was founded by three members and the three founders had 100, 300, and 500 followers, respectively, the company's score would be 300.

As can be seen in **Table 3**, there was a positive correlation between

Table 1
Descriptive statistics for funds raised by companies, age of the company, rounds of investments, and number of board members.

	N	Mean	SD
Cumulative Funds	129	18,120,354.94	29,785,101.64
Funds per year	129	2,125,900.86	3,190,214.28
Age of the Company	129	7.68	5.97
Rounds of Investment	129	2.42	1.71
Number of Board Members	129	1.73	.73

Table 2

Means and standard deviations of founder LinkedIn connections, number of US States founder have worked in, years worked in the industry, and skills of founders.

	N	Mean	SD
LinkedIn Connections	227	1,146.68	1,125.13
Number of US States	227	2.89	2.08
Years Worked in the Industry	227	16.99	8.69
Skills	227	21.69	14.76

founder LinkedIn number of followers and funds raised per year by companies, $r(225) = .29$, $p < .01$. The correlation between the cumulative funds raised by companies and founder LinkedIn followers was $r(225) = .22$, $p < .01$. There are several ways how the social connectedness of a company can be operationalized. The average number of followers controls for differences in the mere number of founders. In addition to using the average number of the followers of a company's founders as an indication of a company's social connectedness, we also considered the sum of all founders' followers, and the maximum number of followers in a company. Consider a company has three founders with 200, 400, and 600 followers, respectively. The mean number of followers in this example would be 400, the sum 1,200, and the maximum 600.

Both, sum of followers, $r(225) = 0.32$, $p < .01$, and founder with the maximum number of followers, $r(225) = 0.31$, $p < .01$, were significantly correlated with the annual funds raised by companies.

As can be seen in Table 3, the cumulative amount of money raised by a company is highly correlated with a company's age. The older the companies, the more money they have raised. Conversely, age of a company was not significantly correlated with the average number of founders' followers on LinkedIn, $r(225) = 0.002$, *ns* (see Table 3).

3.2. Founder experience and startup success

The second hypothesis predicted that the experience of founders, as stated on LinkedIn, will also be related to the funds raised by the company. Again, as funds were collected on the company level while the founder experience was collected at the individual founder level, for the correlation analysis, founder experience was measured as the mean of experience of the founders of a company. In other words, for companies with more than one founder, the average experience of all the founders in a company was used as the measure of experience related to the company. Founder experience was based on three variables—number of skills, number of US states a founder has worked in, and the years a founder has worked in the industry.

As can be seen in Table 3, the average number of skills a company's founders had was significantly correlated with the age of the company ($r(225) = -0.19$, $p < .05$), but it was not significantly correlated with the amount of funds the company raised per year ($r(225) = 0.09$, *ns*).

The average number of years a company's founders have worked in the industry was positively correlated with the funds raised by the

company per year ($r(225) = 0.21$, $p < .01$), age of the company ($r(225) = 0.20$, $p < .01$), average number of contacts founders of a company had on LinkedIn ($r(225) = 0.15$, $p < .01$), the average number of US States founders of a company have worked in ($r(225) = 0.15$, $p < .05$), and the average number of years a founder has worked in the industry ($r(225) = 0.17$, $p < .05$).

The average number of US States a company's founders have worked in did not have a significant correlation with the funds raised by companies per year, $r(225) = -0.075$, *ns*. Speaking to the first research question, as can be seen in Table 3, the mean number of LinkedIn contacts of companies was significantly and positively associated with the experience variables—US States ($r(225) = 0.30$, $p < .01$), skills ($r(225) = 0.82$, $p < .01$), and the number of years founders have worked in the industry ($r(225) = 0.15$, $p < .05$).

3.3. Founder network and experience

To explore our second research question, a multiple regression analysis was conducted with the funds raised per year as the criterion variable and the mean LinkedIn contacts of company founders and the three variables constituting founder experience as predictors. The overall regression equation was significant ($F(4,222) = 10.02$, $p < .01$) with an adjusted R^2 of 0.14. Inspection of the individual variables revealed that the mean of founder contacts of a company was the strongest predictor, ($\beta = 0.35$, $t(226) = 4.99$, $p < .01$). The number of US States founders have worked in ($\beta = -0.19$, $t(226) = -3.04$, $p < .01$) and years founders have worked in the industry ($\beta = 0.19$, $t(226) = 3.13$, $p < .01$) were also significant predictors of funds raised by the company per year. Skills of founders did not significantly improve the prediction of company funds when entered together with the other predictors in a regression analysis ($\beta = -0.07$, $t(226) = -1.05$, *ns*). The strongest predictor of funds raised was the social connectedness of a company, that is, their number of followers. Additional analyses revealed that the social-connectedness cue has a predictive validity of 69%. The predictive validity of a cue is based on an analysis of all logical pairs of choice alternatives in a choice set. The predictive validity is defined as the relative frequency with which a cue makes a correct prediction when looking at all possible pairs of objects for which a cue can be used (Gigerenzer & Todd, 1999; Reimer & Hoffrage, 2006). When analyzing all logical pairs of companies in the current study, the company that had more followers also had higher funds in 69% of the cases in which two companies differed in the number of followers. In other words, using a decision strategy that only capitalizes on social connectedness would yield 69% correct inferences in those situations in which it could be used.

3.4. Differences among types of startup ventures

For our third research question, ANOVAs were conducted to see if there were differences among type of company (operating, acquired, IPO) in the cumulative funds and annual funds raised, the age of companies, and the rounds of investment. There were significant

Table 3

Correlations among funds raised, age of company, mean of founder LinkedIn connections, and mean of founder experience variables.

	Funds per Year	Cumulative Funds	Age of Company	LinkedIn Connections	Number of US States	Skills	Years Worked in Industry
Funds per Year	–	.923**	.86	.29**	-.07	.09	.21**
Cumulative Funds		–	.314**	.22**	-.08	.05	.23**
Age of Company			–	.002	-.19**	-.19**	.20**
LinkedIn Connections				–	.304**	.43**	.15*
Number of US States					–	.14*	.15*
Skills						–	.17*
Years worked in Industry							–

**Correlation is significant at the 0.01 level.

*Correlation is significant at the 0.05 level.

differences in all four variables (cumulative funds: $F(2,224) = 45.04$, $p < .01$; funds per year: $F(2,224) = 26.42$, $p < .01$; age of company: $F(2,224) = 40.45$, $p < .01$; rounds of investment: $F(2,224) = 33.49$, $p < .01$). IPOs raised the most amount of money ($M = 54$ million; $SD = 45$ million), followed by acquired ($M = 15$ million, $SD = 25$ million), and operating companies respectively ($M = 5.8$ million, $SD = 10$ million).

Another ANOVA was conducted to explore if the three types of companies differed systematically in founder LinkedIn followers and experience attributes. The two significant differences among the three types of companies referred to the number of years a founder has worked in the industry, $F(2,224) = 3.58$, $p < .05$, and the average number of LinkedIn followers that founders in a company had, $F(2,224) = 5.8$, $p < .05$. Founders of IPOs had worked more years ($M = 19.9$, $SD = 13.1$) in the industry than founders of operating company ($M = 15.4$, $SD = 6.9$). On average, founders in companies that were IPO had more LinkedIn followers ($M = 1,531.13$, $SD = 1,047.16$) followed by acquired companies ($M = 1,072.84$, $SD = 902.54$) and operation companies ($M = 1,006.63$, $SD = 794.93$). The differences in the remaining variables were not significant.

In the light of these differences, we also conducted bivariate correlations between the mean of founder followers in companies and the amount of annual funds raised by companies separately for each of the three kinds of companies. This analysis revealed that the number of founder followers was not significantly correlated with annual funds within any of the three categories: IPO companies ($r(27) = 0.28$, *ns.*), operating companies ($r(47) = -0.03$, *ns.*), and acquired companies ($r(47) = -0.09$, *ns.*). The companies in the three categories of IPO, acquired and operating categories were more homogenous in terms of funds raised, number of founders, and age than the overall sample.

4. Discussion

This study looked at whether social networks of startup founders are related to tangible financial outcomes of a startup. Breaking away from the tradition of relying on self-reported ego centered network data, this study utilized social networking site LinkedIn to acquire data on network size and experience of founders. As expected, social networks of startup founders were associated with tangible financial outcomes of companies. Startup founders with more LinkedIn followers had raised more money for their respective companies. Moreover, founder experience was also correlated with a startup's ability to raise funds. Founders who have worked in the industry for longer are also those who were more successful in raising funds for their companies.

In addition, one of the major takeaways of this study is that social connectedness of a founder is the best predictor of funds raised annually by a founder. Moreover, skills of a founder did not significantly increase the predictability of company funds. We found that the predictive validity of the social connectedness cue is 69%. This means, when analyzing all pairs of companies in the current study, the company that had more followers also had higher funds in 69% of the cases in which two companies differed in the number of followers. A 69% predictive validity or accuracy based on one cue is substantial compared to other information-intensive decision-making models that include more cues, including financial parameters, to predict a start-up's ability to raise investment funds (see Roure & Keeley, 1990; Zacharakis & Meyer, 2000).

What does this mean for founders? The current study provides evidence that data on social networking site LinkedIn is associated with how much money companies raise. Moreover, it is the strongest predictor of startup annual funds. Correlational studies like this study do not test causal relationships between variables, and, thus, it cannot be concluded from this study that the size of social networks affect funding. However, at the same time, the observed correlations are in accordance with and do not speak against the conventional wisdom that tells founders to invest time and energy into networking can potentially

assist in fundraising efforts. The findings are in accordance with the notion that it is not only important what you know but also who and how many people you know.

Moreover, this study also shows that while LinkedIn networks might represent a limited snapshot of a founder's network, the network is associated with the amount of money a company is able to raise. It also indicates that founders can potentially use the platform to not only build their social connections but communicate the worth of their connectedness to investors.

There are, however, several limitations of this study that can be explored through further research. First, this study looks at only one dimension of founder networks, which is the network size. This study focusses solely on the quantity of LinkedIn connections of founders. This is one of the limitations of using an online resource for analysis. However, there are issues with self-reported ego network data and online resources provide a viable alternative resource to study social networks and their association with tangible organizational outcomes. Future research should investigate other structural properties of founder networks and their relationship with funds raised by companies. Second, this study does not provide an answer to the question whether more connections lead to more funds or vice versa as the study is merely correlational. Future research can look at longitudinal analysis of founder connections and funds raised. Another limitation of this study is the reliability of data that online resources like LinkedIn and Crunchbase provide. However, online resources do represent the current entrepreneurial environment where communication about networks are often conducted through online platforms. Moreover, given the lack of transparency of financial information available for various startup organizations, crowdsourced information sources like Crunchbase provide more extensive data than other information sources such as self-reports in surveys. However, it would be useful to test the network hypothesis in future studies with data from other information sources to explore if the findings from the current study can be replicated across different data sources that describe the funding of startup companies and the social networks of their founders.

Another limitation that warrants further study are the differences among the three kinds of startup companies that were used in this study. The positive correlation between annual funds and founder followers were not significant when the three categories of companies (operating, acquired and IPO companies) were analyzed separately. This can be explained in part by the observation that variation in funds raised within the categories was much smaller than across categories. It would be worthwhile, in future research, to look into a longitudinal analysis that covers a journey of a startup firm from operating to IPO which is likely to provide a better understanding of the role and the trajectory of connections, funding and other variables that companies have in differing stages of startup development.

To conclude, the major contribution of this study is that it extends the study of online social networks to the entrepreneurial context and investment decision-making by looking at the relationship between founder LinkedIn connections, founder experience, and funds raised by a company. The study found that founder connectedness was a better predictor of company investment funds than founder experience variables. This study, therefore, provides some empirical support for the idea that it matters more “who you know than what you know.”

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